Species-Aware Review: Bacterial Non-Homologous End Joining (Ku–LigD) in *Pseudomonas putida* KT2440

Module/bucket: kegg:ppu03450 — Non-homologous end-joining **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module area:** other_kegg_pathway | **Candidate genes:** `ku` (PP_3255 / Q88HU8), `ligD` (PP_3260 / Q88HU3)

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1. Executive summary

The bacterial NHEJ module is **satisfied in *P. putida* KT2440 by exactly the two candidate genes**, Ku (PP_3255) and LigD (PP_3260), and this is supported by **direct experimental evidence in the target strain** — an unusually strong footing for a bacterial NHEJ call. Genetic ablation of either *ku* or *ligD* alters DSB-repair/mutagenesis phenotypes in KT2440 (Paris et al. 2015,

P 25942369

; Sharaev et al. 2022,

P 36475478

). All three mechanistic sub-steps of the module (Ku-dependent end recognition/synapsis; LigD polymerase gap-fill and 3′-phosphoesterase end-healing; LigD ATP-dependent sealing) map to just these two proteins, with LigD being a single multifunctional polypeptide (POL + PE + LIG).

Key curation cautions: (i) **EC 6.5.1.1 captures only the ligase activity of LigD** — its polymerase and phosphoesterase activities need separate GO terms and are not conveyed by the enzyme name/EC; (ii) the gene symbol **"ligC" in KT2440 (PP_2602) is a homonym false-friend** — it is 4-carboxy-2-hydroxymuconate-6-semialdehyde dehydrogenase (aromatic catabolism, EC 1.1.1.312), **not** a backup NHEJ ligase; (iii) the **mycobacterial-style LigC/PrimC backup NHEJ routes are absent** and should be marked not_expected_in_target_taxon. Both candidate genes merit promotion to full `fetch-gene` review, LigD especially because of its multi-activity/single-EC mismatch.

2. Target-organism pathway definition

Included process. The Ku–LigD "two-component" bacterial NHEJ pathway: template-independent repair of a chromosomal (or plasmid) DNA **double-strand break (DSB)** in which (1) a **Ku homodimer** binds, protects, and juxtaposes the two broken duplex ends and recruits LigD; (2) **LigD** conditionally remodels non-ligatable termini via a **polymerase (POL)** module (gap fill, with a marked preference for ribonucleotide addition) and a **3′-phosphoesterase (PE)** module (removes 3′-phosphate/3′-blocking groups to yield a ligatable 3′-OH); and (3) the **ATP-dependent ligase (LIG)** module seals the backbone. The pathway is characteristically **error-prone/mutagenic** and is most important in stationary phase / quiescent, non-replicating cells where a sister chromosome for homologous recombination (HR) is unavailable.

Kept separate (neighboring processes / overview maps). - **Homologous recombination** (RecA/RecBCD/AddAB; KEGG "Homologous recombination" ppu03440) — the alternative DSB pathway that requires a template; explicitly excluded. - **Base excision repair / mismatch repair / nucleotide excision repair** overview maps — even though Ku has a reported AP/5′-dRP-lyase moonlighting activity (P 25355514), BER proper is a separate module. - **NAD$^+$-dependent replicative/backup ligases** LigA (PP_4274, EC 6.5.1.2) and LigB (PP_4968, EC 6.5.1.2) — these seal nicks in replication/repair but are **not** the ATP-dependent NHEJ ligase and are not part of this module. - **Eukaryotic NHEJ factors** (DNA-PKcs, XRCC4, LIG4, Artemis, etc.) — not present in bacteria; excluded.

Alternate names / database definitions. "Bacterial/prokaryotic NHEJ"; "Ku–LigD end-joining"; KEGG map03450 (organism-specific ppu03450). LigD is also called "DNA ligase D," "NHEJ DNA ligase/polymerase," "LigD (POL/PE/LIG)." Ku is "Non-homologous end joining protein Ku," "prokaryotic Ku," "bacterial Ku."

3. Expected step model (module satisfiability)

Module step	Expected player	KT2440 gene (domain)	Status	Evidence strength
1. DNA-end recognition, protection & synapsis	prokaryotic Ku (dsDNA-end binding)	**ku / PP_3255 (Q88HU8)** — Ku β-barrel ~11–194	**covered**	Direct (target strain, genetics) + genus biochemistry + InterPro domain
2a. Gap-filling polymerase	LigD POL (DNA/ RNA polymerase)	**ligD / PP_3260 (Q88HU3)** — POL ~547–812	**covered**	Direct (target strain: POL domain implicated) + genus structure + InterPro (PaeLigD-type)
2b. 3′-end healing	LigD PE (3′-phosphoesterase)	**ligD / PP_3260** — PE ~5–160	**covered**	Direct (target strain: PE domain implicated) + InterPro (TIGR02777)
3. ATP-dependent sealing	LigD LIG (ATP ligase, EC 6.5.1.1)	**ligD / PP_3260** — LIG ~219–520	**covered**	Direct (UniProt EC 6.5.1.1) + genus end-joining assays + InterPro (TIGR02779)
(Backup) LigD-	mycobacterial LigC / PrimC	*none* (PP_2602	**not_expected_in_target_taxon**	Bioinformatic (proteome

Module step	Expected player	KT2440 gene (domain)	Status	Evidence strength
independent ligase		"ligC" is unrelated)		scan) + comparative

All four core steps are marked `covered`. No core step is a gap. The only "missing" element relative to the fully generic bacterial-DSB picture is a dedicated backup ligase, which is a genuine biological absence in this lineage rather than an annotation gap.

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4. Candidate genes and evidence

4.1 `ku` — PP_3255 / Q88HU8 (Non-homologous end joining protein Ku)

- **Role:** DNA-end-binding homodimer; recognizes and protects duplex DNA ends, promotes synapsis, and recruits/stimulates LigD.
- **Evidence type:** *Direct target-strain genetics* — Δ*ku* changes the stationary-phase mutation spectrum in carbon-starved KT2440 (P 25942369) and alters Cas9-DSB repair outcomes in KT2440 (P 36475478). *Genus biochemistry* — Pseudomonas Ku stimulates LigD POL activity and promotes full-length LigD end-joining (P 20018881). *Structural model* — cryo-EM of *M. tuberculosis* Ku shows homodimer→proteofilament on DNA and a C-terminus that regulates DNA loading and LigD recruitment (P 41298423; transfer to KT2440 is moderate — conserved fold, different species).
- **UniProt/InterPro features:** 273 aa; prokaryotic-type Ku domain (CDD cd00789 KU_like; Pfam PF02735 Ku70/Ku80 β-barrel ~11–194; IPR009187 prokaryotic Ku; HAMAP MF_01875), with a **C-terminal extension (~195–273)** beyond the core β-barrel that corresponds to the

region implicated in DNA loading/LigD recruitment (cf. *P. aeruginosa* Ku C-terminal truncations mapping the functional core to Ku-(1–229),

P 20018881 ;

and the C-terminal regulatory role in *M. tuberculosis* Ku,

P 41298423).

Single-copy — **no Ku paralog** in the proteome scan.

- **Curation caveats:** GO "double-stranded DNA binding" and "DSB repair via NHEJ" are well supported. A reported AP/5′-dRP-lyase side activity of bacterial Ku (shown for *B. subtilis* and *P. aeruginosa* Ku,

  P 25355514)

  is a *moonlighting* function; do **not** promote it to a core NHEJ annotation for KT2440 without direct data.

4.2 `ligD` — PP_3260 / Q88HU3 (DNA ligase D; EC 6.5.1.1)

- **Role:** Multifunctional NHEJ end-processing/sealing enzyme carrying three autonomous modules — POL (gap fill, rNTP-preferring, Mn^{2+}-dependent), PE (3′-phosphoesterase → 3′-OH), and LIG (ATP-dependent sealing).
- **Evidence type:** *Direct target-strain genetics* — Δ*ligD* alters the KT2440 stationary-phase mutation spectrum, and **both PE and POL domains are implicated** in the mutation phenotype

  (P 25942369);

  KT2440 Cas9-DSB repair depends on LigD

  (P 36475478).

  Genus structure/biochemistry — 1.5 Å crystal structure of the *Pseudomonas* LigD POL domain (primase-superfamily, two-metal, rNTP preference;

  P 16446439)

  and detailed gap-fill/Ku-stimulation biochemistry

  (P 20018881).

  Domain framework — LigD three-module review

  (P 34901162).

- **UniProt features:** 833 aa; annotated "ATP-dependent DNA ligase family profile" (307–399); keywords include **DNA-directed DNA polymerase, Exonuclease/Nuclease, Manganese, Nucleotidyltransferase, Multifunctional enzyme** — consistent with a full POL+PE+LIG LigD.
- **InterPro domain architecture (curation-ready boundaries; order N→C is PE–LIG–POL):**
- **PE (3′-phosphoesterase)** ~residues **5–160** — TIGR02777 / IPR014144 "DNA ligase D, 3′-phosphoesterase domain"; Pfam PF13298 (38–143). → GO: polynucleotide 3′-phosphatase activity (GO:0046403).
- **LIG (ATP-dependent ligase/adenylation)** ~residues **219–520** — TIGR02779 / IPR014146; Pfam PF01068 (adenylation, 219–399) + PF04679 (C-terminal, 418–514) + OB-fold (IPR012340, 401–522). → GO: DNA ligase (ATP) activity (GO:0003910); this is the sole activity covered by EC 6.5.1.1.
- **POL (primase-superfamily polymerase)** ~residues **547–812** — TIGR02778 / IPR014145; Pfam PF21686; CDD cd04862 **"PaeLigD_Pol_like"**; IPR033651 **"LigD polymerase domain, PaeLigD-type"**. → GO: DNA-directed DNA polymerase activity (GO:0003887).
- Overall: IPR014143 "DNA ligase D" (228–816); NCBIfam TIGR02776; PANTHER PTHR42705. The POL module being explicitly classified as **PaeLigD-type** ties KT2440 LigD directly to the *P. aeruginosa* LigD Pol structure (P 16446439), making genus→KT2440 mechanistic transfer **strong** at the fold level.
- **Curation caveats (important):**
- **Single-EC / multi-activity mismatch:** the recommended EC (6.5.1.1) and the "DNA ligase (ATP)" name describe **only** the LIG module. The POL and PE activities are biologically essential module steps but are invisible to the EC/name. Ensure GO annotations include **DNA-directed DNA polymerase activity** (GO:0003887 / or DNA/RNA polymerase) and **polynucleotide 3′-phosphatase activity** (GO:0046403) in addition to **DNA ligase (ATP) activity** (GO:0003910).
- **Over-propagation risk in reverse:** automated pipelines keyed only on "DNA ligase (ATP)" may under-annotate LigD (missing POL/PE). Conversely, generic "DNA ligase" text should not be conflated with the NAD^+ ligases LigA/LigB.
- No KT2440-specific crystal structure exists; atomic data are genus-level (mostly *P. aeruginosa*). Ortholog transfer is **strong** but should be flagged as inferred at the structural level.

4.3 Genomic context

`ku` (PP_3255) and `ligD` (PP_3260) sit in the same chromosomal region but are **not part of a shared operon**: the four intervening genes are functionally unrelated to DNA repair — PP_3256 (glycosyltransferase, group 2), PP_3257 (methyltransferase), PP_3258 (acetylglucosaminylphosphatidylinositol deacetylase), and PP_3259 (acyl-CoA dehydrogenase-related). Thus *ku* and *ligD* are **independently encoded and likely independently regulated**, consistent with the genetic observation that Ku and LigD can act in partly separate mutagenic sub-pathways in KT2440 (Paris et al. 2015, P 25942369). Unlike some bacteria where *ku* and *ligD* are adjacent/co-transcribed, this locus arrangement argues against inferring co-regulation, and any module reconstruction should treat the two as separate transcriptional units absent operon/transcriptomic data.

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5. Gaps, ambiguities, and likely over-annotations

1. **"ligC" (PP_2602 / Q88JP7) is NOT an NHEJ ligase.** It is 4-carboxy-2-hydroxymuconate-6-semialdehyde dehydrogenase (EC 1.1.1.312), a protocatechuate/aromatic-degradation enzyme. This is a classic gene-symbol homonym; any mapping of KT2440 "ligC" to bacterial NHEJ backup would be an **over-annotation error**. Mark the LigC/PrimC backup steps `not_expected_in_target_taxon`.
2. **No dedicated backup NHEJ ligase.** Proteome scan shows LigD is the **sole ATP-dependent DNA ligase (EC 6.5.1.1)** in KT2440; LigA (PP_4274) and LigB (PP_4968) are NAD^+-dependent (EC 6.5.1.2). Mycobacteria-style LigC/PrimC redundancy is a lineage-specific feature, not general.
3. **LigD-independent NHEJ caveat (uncertain).** A 2026 study reports that *P. aeruginosa* can perform **LigD-independent NHEJ** and microhomology-mediated repair, with Ku (but not LigD) conditionally essential in some contexts (P 42306942). The responsible ligase is not defined and this is a *related-species* result. It does **not** change the KT2440 module core, but flags that an as-yet-unidentified alternative sealing route may exist in pseudomonads — an open question, not a curatable gene.

4. **Ku moonlighting (AP-lyase).** Do not over-extend Ku's annotation to BER on the basis of the *B. subtilis*/*P. aeruginosa* lyase activity (P 25355514) without direct KT2440 evidence.
5. **Species transfer bookkeeping.** Distinguish: KT2440-direct (P 25942369 P 36475478) > genus *Pseudomonas* biochemistry/structure (P 16446439 P 20018881) > generic bacterial/mycobacterial mechanism (P 34901162 P 15778718 P 24957619 P 41298423).

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6. Module and GO-curation recommendations

- **Step calls:** end recognition/synapsis → **covered** (Ku); gap-fill polymerase → **covered** (LigD POL); 3′-end healing → **covered** (LigD PE); ATP-dependent sealing → **covered** (LigD LIG). Backup ligase → **not_expected_in_target_taxon**.
- **Module boundaries:** the generic module scope (Ku–LigD core, excluding HR/SSA/eukaryotic NHEJ/backup routes) is **correct for KT2440**. No `module_needs_revision` required; the "conditional" remodeling framing matches the observed data (Ku alone can present compatible ends for direct sealing; POL/PE act only when termini are non-ligatable).
- **GO term needs for LigD (Q88HU3):** ensure co-annotation of (a) DNA ligase (ATP) activity GO:0003910; (b) DNA-directed DNA polymerase / DNA polymerase activity (POL module); (c)

polynucleotide 3′-phosphatase activity GO:0046403 (PE module); and process term DSB repair via NHEJ GO:0006303. Flag the EC-vs-activity mismatch so downstream EC→GO pipelines do not drop POL/PE.
- **GO for Ku (Q88HU8):** double-stranded DNA binding GO:0003690 (or dsDNA end binding), DSB repair via NHEJ GO:0006303.
- **No new module document** appears necessary; the existing generic bacterial-NHEJ module applies cleanly. A curator note recording the "ligC" homonym and the LigD single-EC/multi-activity issue is recommended.

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7. Genes to promote to full `fetch-gene` review

1. `ligD` / **PP_3260 (Q88HU3)** — **High priority.** Multifunctional enzyme whose EC/name conveys only 1 of 3 activities; verify POL and PE module annotations and GO coverage; confirm domain boundaries against AlphaFold/InterPro.
2. `ku` / **PP_3255 (Q88HU8)** — **Medium priority.** Straightforward core call; confirm single-copy status and check whether to note (without promoting) the AP-lyase moonlighting literature.
3. **(Watchlist, not a candidate)** `ligC` / **PP_2602 (Q88JP7)** — flag explicitly as a **name collision** to prevent erroneous NHEJ mapping; no NHEJ review needed.

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8. Key references

- Paris Q, et al. *NHEJ enzymes LigD and Ku participate in stationary-phase mutagenesis in Pseudomonas putida.* 2015.

  P 25942369

  — direct KT2440/putida genetics; PE+POL domains implicated.
- Sharaev N, et al. *Repair of Double-Stranded DNA Breaks Generated by CRISPR-Cas9 in Pseudomonas putida KT2440.* 2022.

  P 36475478

  — direct KT2440 NHEJ manipulation.

- Zhu H, et al. *Atomic structure and NHEJ function of the polymerase component of bacterial DNA ligase D.* 2006.

  P 16446439

  — *Pseudomonas LigD POL crystal structure.*
- Zhu H, Shuman S. *Gap filling activities of Pseudomonas LigD polymerase and functional interactions with Ku.* 2010.

  P 20018881

  — *Pseudomonas Ku–LigD biochemistry.*
- Amare B, et al. *LigD: A Structural Guide to the Multi-Tool of Bacterial NHEJ.* 2021.

  P 34901162

  — *POL/PE/LIG module review.*
- de Ory A, et al. *Efficient processing of abasic sites by bacterial NHEJ Ku proteins.* 2014.

  P 25355514

  — *Ku AP/5′-dRP-lyase moonlighting (Bsu, Pae).*
- Bhattarai H, et al. *DNA ligase C1 mediates LigD-independent NHEJ of M. smegmatis.* 2014.

  P 24957619

  — *LigC backup (mycobacteria; absent in Pseudomonas).*
- Gong C, et al. *Mechanism of NHEJ in mycobacteria: Ku, LigD and LigC.* 2005.

  P 15778718

  — *backup LigC concept.*
- Zahid S, et al. *Oligomerisation of Ku from M. tuberculosis promotes DNA synapsis.* 2025.

  P 41298423

  — *Ku structure/synapsis (cross-species).*
- Hare PJ, et al. *Non-canonical DSB repair in P. aeruginosa includes LigD-independent NHEJ.* 2026.

  P 42306942

  — *open question on alternative routes (related species).*

Uncertainty ledger

- **Direct (KT2440):** functional requirement of Ku and LigD, and involvement of LigD POL+PE domains, in DSB repair/mutagenesis.
- **Strong inference (genus Pseudomonas):** LigD POL structure/mechanism, Ku–LigD stimulation, ribonucleotide-preferring gap fill.
- **Comparative/generic:** three-module LigD architecture, PE→3′-OH chemistry, Ku synapsis mechanics, absence of LigC/PrimC backup (bioinformatic for KT2440).
- **Open:** existence/identity of any LigD-independent sealing route in pseudomonads; in-vivo relevance of Ku AP-lyase in KT2440; operon structure of ku/ligD.
